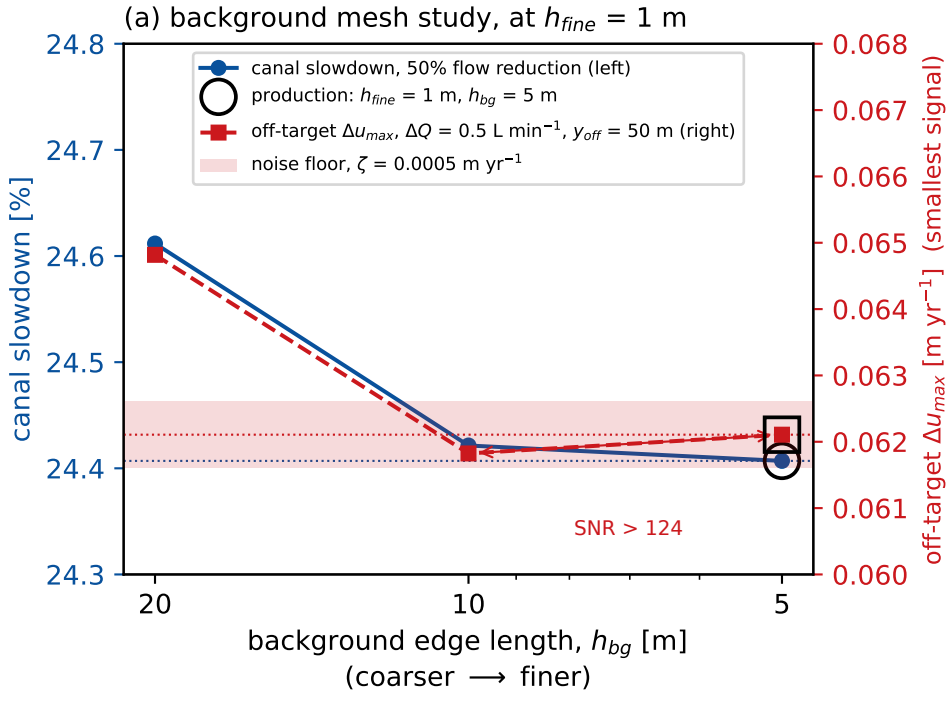
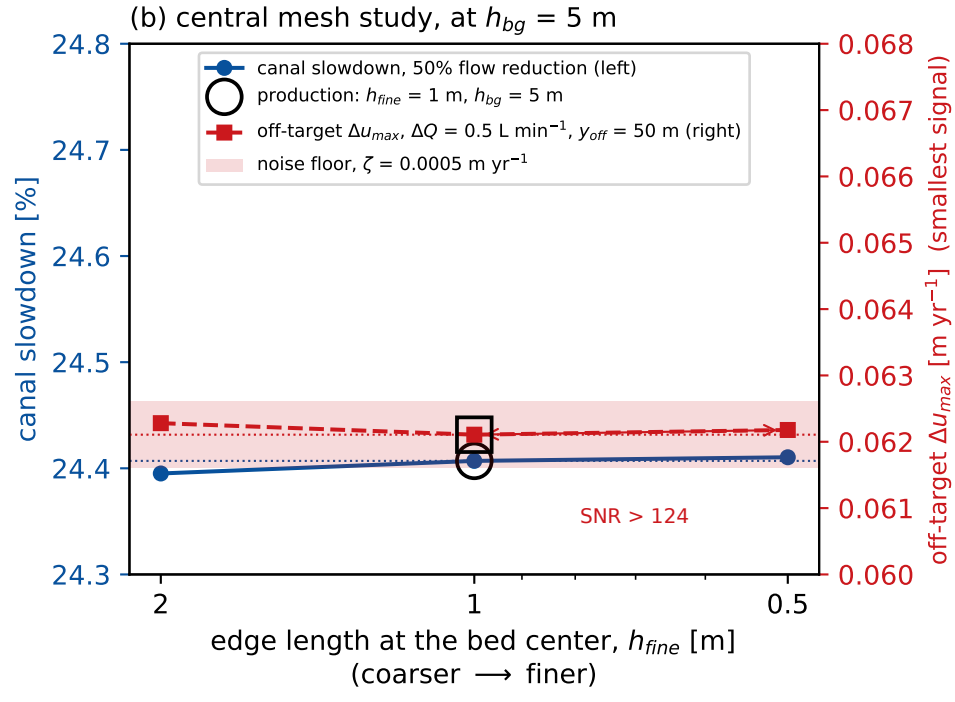
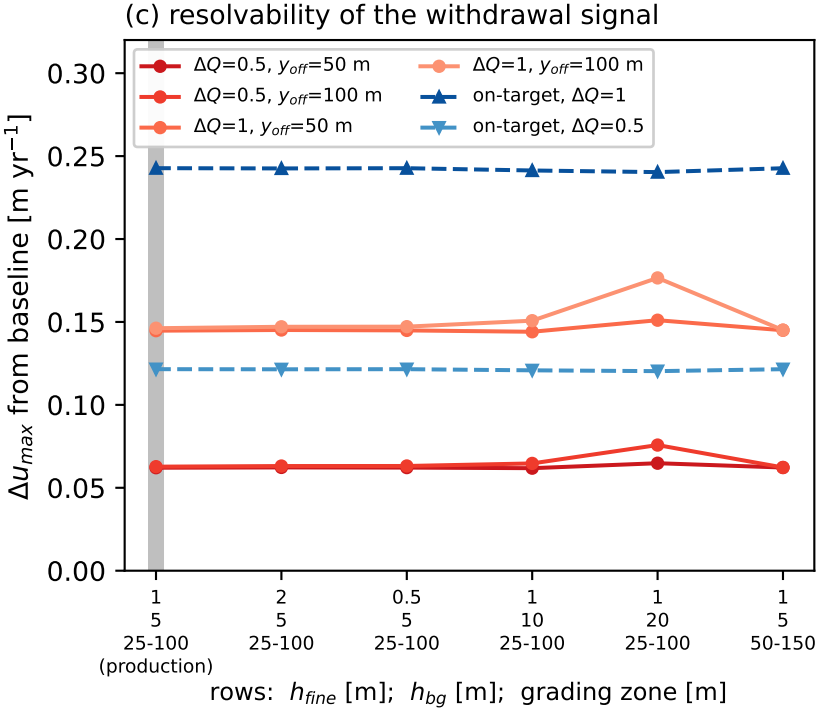
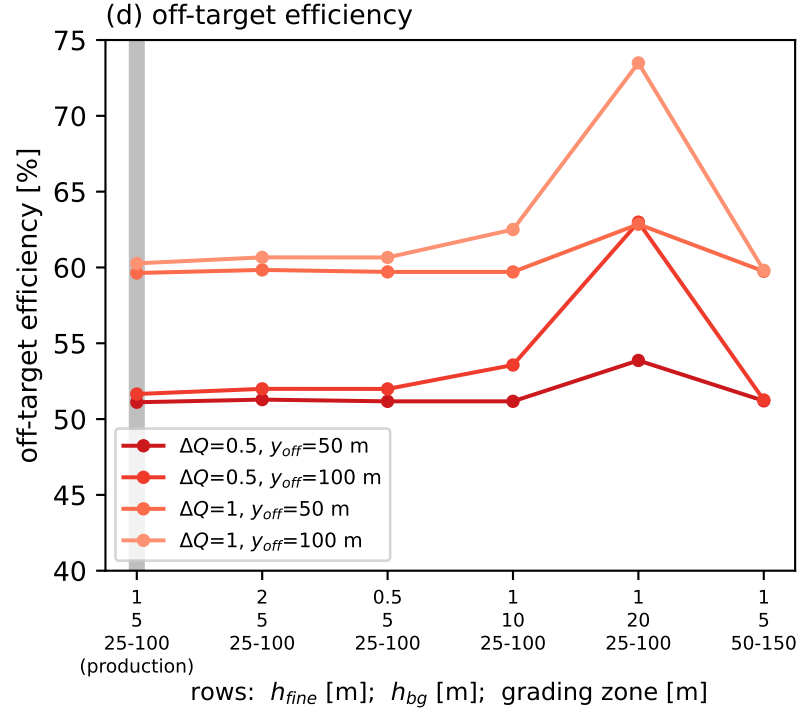

(a) background mesh study, at h_{fine} = 1 m
canal slowdown, 50% flow reduction (left)
production: h_{fine} = 1 m, h_{bg} = 5 m
off-target Δu_{max}, ΔQ = 0.5 L min^{-1}, y_{off} = 50 m (right)
noise floor, ζ = 0.0005 m yr^{-1}
SNR > 124
canal slowdown [%]
off-target Δu_{max} [m yr^{-1}] (smallest signal)
background edge length, h_{bg} [m]
(coarser ⟶ finer)


(b) central mesh study, at h_{bg} = 5 m
canal slowdown, 50% flow reduction (left)
production: h_{fine} = 1 m, h_{bg} = 5 m
off-target Δu_{max}, ΔQ = 0.5 L min^{-1}, y_{off} = 50 m (right)
noise floor, ζ = 0.0005 m yr^{-1}
SNR > 124
canal slowdown [%]
off-target Δu_{max} [m yr^{-1}] (smallest signal)
edge length at the bed center, h_{fine} [m]
(coarser ⟶ finer)


(c) resolvability of the withdrawal signal
ΔQ=0.5, y_{off}=50 m
ΔQ=0.5, y_{off}=100 m
ΔQ=1, y_{off}=50 m
ΔQ=1, y_{off}=100 m
on-target, ΔQ=1
on-target, ΔQ=0.5
Δu_{max} from baseline [m yr^{-1}]
rows: h_{fine} [m]; h_{bg} [m]; grading zone [m]


(d) off-target efficiency
ΔQ=0.5, y_{off}=50 m
ΔQ=0.5, y_{off}=100 m
ΔQ=1, y_{off}=50 m
ΔQ=1, y_{off}=100 m
off-target efficiency [%]
rows: h_{fine} [m]; h_{bg} [m]; grading zone [m]